

This addendum written by Sergey Glazer

See complete discussions on the class at

https://groups.io/g/cdl/topic/class_in_basic_electronics/112405164

This is an addendum to Joe Wilkes' [presentation](#) for [CDL's](#) Basic Electronics class in June, 2025. The purpose of this is to provide a more detailed explanation to derivation of the formulas on slides 32 and 33. Information copied from these slides is in **bold**.

The goal is to understand the stability of voltages in this circuit relative to variation in the gain of transistor Q_1 . We are analyzing only the DC behavior. We want to compute voltages at each connection of Q_1 for a range of values of gain, so we can plot them and examine how they vary.

Slide 32 of the presentation includes the initial conditions:

- **$R_8 = 2.2 \text{ k}\Omega$** , resistor between emitter and ground
- **$R_6 = 6.8 \text{ k}\Omega$** , resistor between V_1 and base
- **$R_7 = 4.7 \text{ k}\Omega$** , resistor between V_2 and collector
- **$V_1 = 1.6 \text{ V}$** , voltage applied to the base circuit, as produced by the drop of the LED
- **$V_2 = 9 \text{ V}$** , voltage applied to the collector circuit, as produced by the battery (we are ignoring R_0 : it's assumed to have very low resistance)

Continuing on slide 33, the voltages on each side of Q_1 are:

1. **$V_E = R_8 I_E$** , Ohm's law for voltage/current over R_8
 - a. **$= R_8 (I_B + I_C)$** , Kirchhoff's current law, assuming no losses in Q_1
2. **$V_B = V_E + 0.7$** , since the voltage drop by the B-E junction of Q_1 is 0.7 V
3. **$V_C = V_2 - I_C R_8$** , supply voltage less the drop over R_8

And the corresponding currents are:

4. **$I_E = V_E / R_8$**
5. **$I_B = (V_1 - 0.7 - V_E) / R_6$** because $V_1 - V_B$ is the voltage drop over R_6
6. **$I_C = G I_B$** , where G is the gain of Q_1

Now we can derive I_B based on known values:

$$I_B = (V_1 - 0.7 - R_8(I_B + I_C)) / R_6 \text{ from 5 and 1a}$$

$$I_B R_6 = V_1 - 0.7 - R_8 I_B - R_8 G I_B \text{ from above and 6}$$

$$I_B R_6 + I_B R_8 + I_B G R_8 = V_1 - 0.7 \text{ collecting terms with } I_B$$

$$\mathbf{I_B = (V_1 - 0.7) / (R_6 + R_8(G + 1))}$$

Now we can compute I_C using 6 and compute I_E as the sum of I_B and I_C .

From these currents we can compute the voltages at all three Q_1 connections using 1-3.

These calculations are implemented in the spreadsheet on slide 34.